Relatives' similarity in test score performance is explained by a wide range of genetic and environmental parameters

Supplemental slides

Nuno R. G. Carvalho[1,2] and Alexander "Sasha" Gusev[1,2]

[1]Harvard Medical School [2]Dana-Farber Cancer Institute

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Reanalysis of Eftedal NH, Eilertsen EM, Sunde HF, Kleppestø TH, Ystrom E, Czajkowski NO (2025). *PNAS* 122(25):e2419627122.

github.com/nunorgcarvalho/eftedal_reanalysis nuno_ramosgrangeiacarvalho@g.harvard.edu

The core models

how to read the model slides

Components

- **A**: additive genetic effects under assortative mating (AM); the model of Eftedal et al.
- **GxG**: pairwise gene–gene interactions, decaying as $(1/4)^n$ with genetic degree n (not affected by AM).
- **C**: shared environment, $V_C\, c^f$ with family degree f; split into a within-sibling part $V_C(1-c)$ and a beyond-sibling part $V_C\, c$.

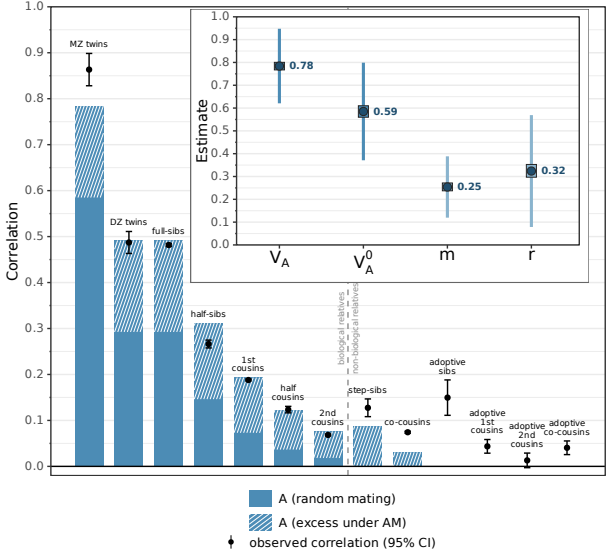
Naming

- **+** : components fitted jointly.
- $\rightarrow$: fitted in steps. The components before the arrow are fitted first and their parameters are then held fixed while those after the arrow are fitted.

Model slides

Revised estimator, optimized for the Marginal Type, as on the poster. Marginal Pair CIs: parametric bootstrap; Marginal Type CIs: leave-one-type-out jackknife. CV: leave-one-type-out cross-validation (RMSE; rank among the nine models, 1 = best).

symbol	meaning
Additive model	
n	genetic degree
V_A	additive genetic variance
V_A^0	additive variance under random mating, $V_A(1-m)$
m	mates' genetic correlation
r	mates' phenotypic correlation
Gene–gene interactions	
V_{GxG}	GxG variance
Shared environment	
V_C	shared-environment variance
f	family degree
c	decay per family degree



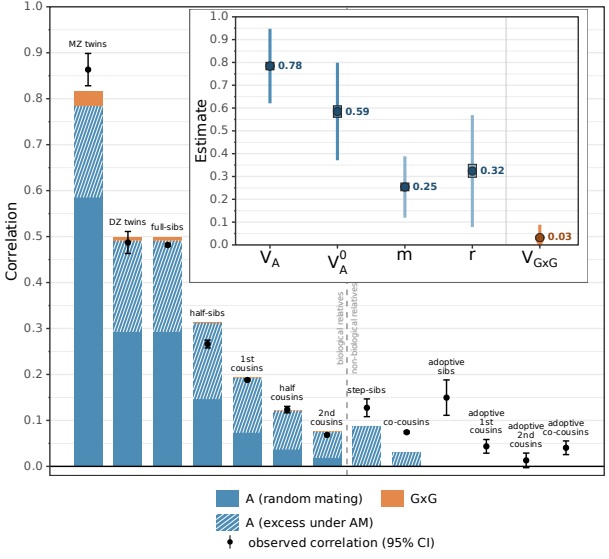
Model

Additive genetic effects under assortative mating only: the model of Eftedal et al., fitted with the revised estimator.

Fit (Marginal Type)

χ^2/df	28.1
CV RMSE	0.070
CV rank	7 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



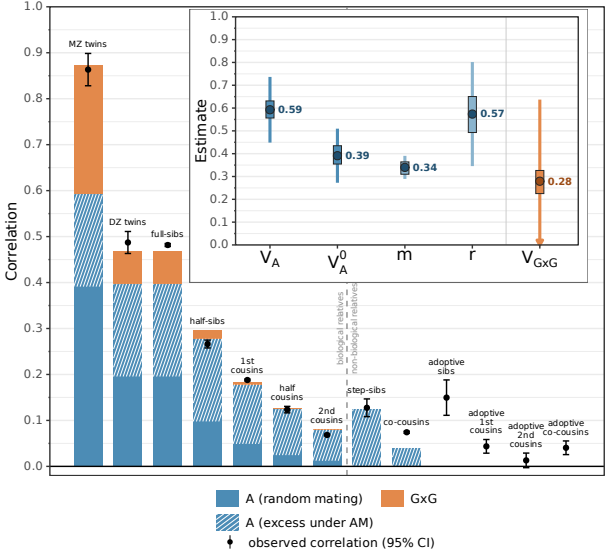
Model

A fitted first; **GxG** then fitted to what remains, with **A**'s parameters held fixed.

Fit (Marginal Type)

χ^2/df	29.5
CV RMSE	0.065
CV rank	5 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



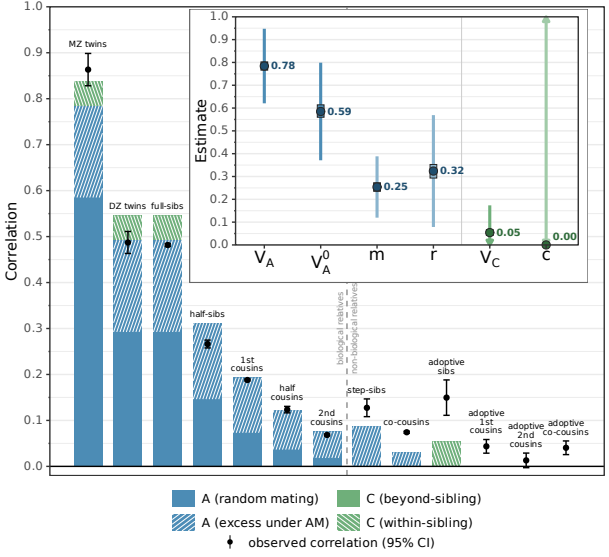
Model

A and **GxG** fitted jointly.

Fit (Marginal Type)

χ^2/df	21.2
CV RMSE	0.063
CV rank	4 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



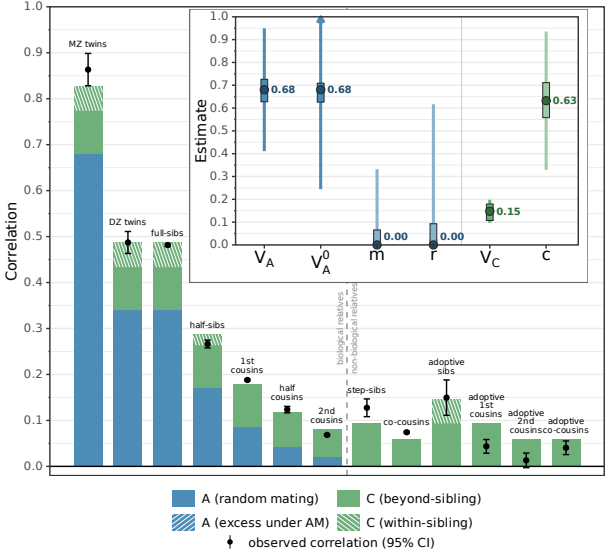
Model

A fitted first; **C** then fitted to what remains, with **A**'s parameters held fixed.

Fit (Marginal Type)

χ^2/df	47.7
CV RMSE	0.071
CV rank	8 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



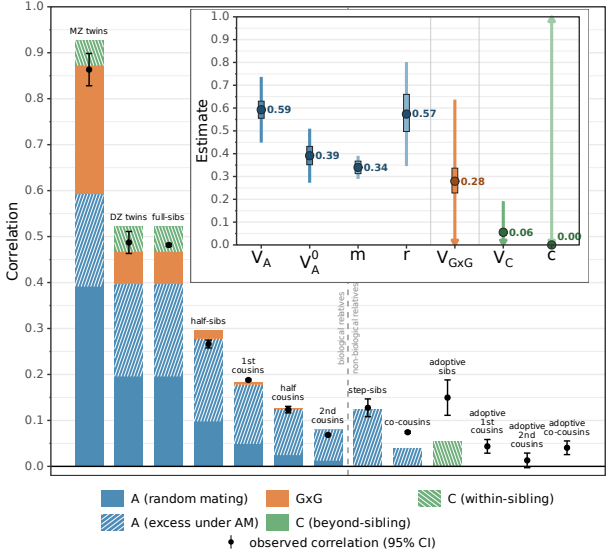
Model

A and **C** fitted jointly.

Fit (Marginal Type)

χ^2/df	11.5
CV RMSE	0.050
CV rank	3 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



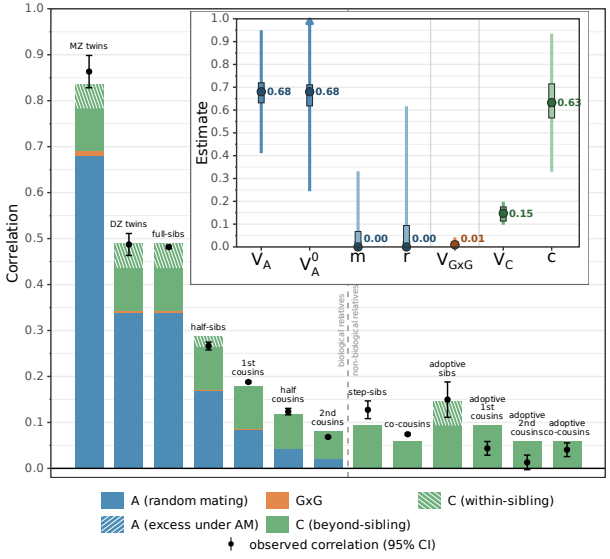
Model

A and **GxG** fitted jointly first; **C** then fitted with their parameters held fixed.

Fit (Marginal Type)

χ^2/df	28.9
CV RMSE	0.065
CV rank	6 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



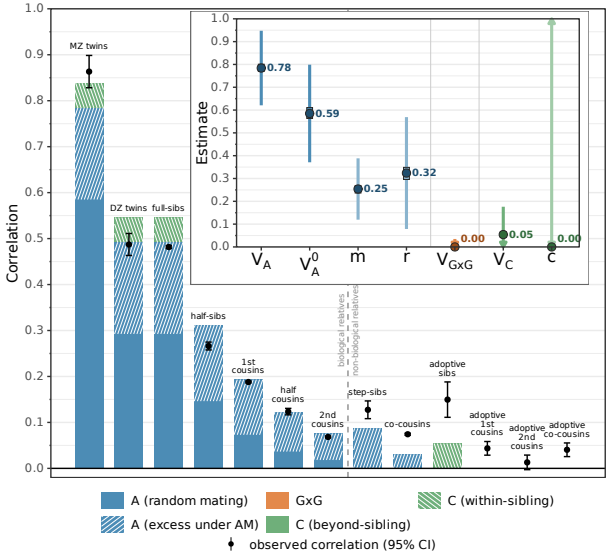
Model

A and **C** fitted jointly first; **GxG** then fitted with their parameters held fixed.

Fit (Marginal Type)

χ^2/df	11.7
CV RMSE	0.046
CV rank	2 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



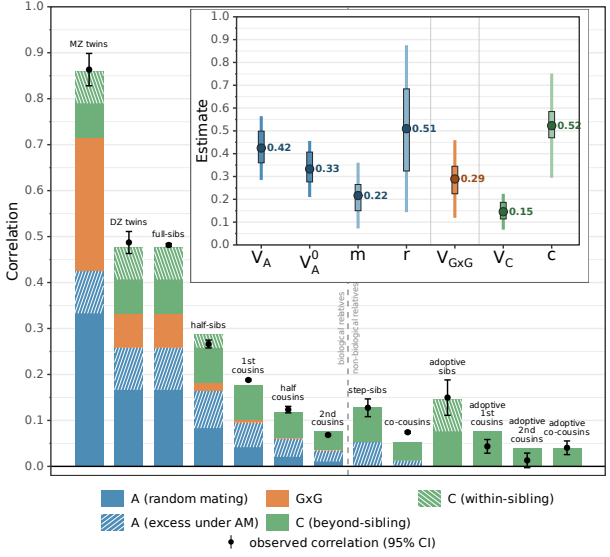
Model

A fitted first; **GxG** and **C** then fitted jointly, with **A**'s parameters held fixed.

Fit (Marginal Type)

χ^2/df	48.3
CV RMSE	0.072
CV rank	9 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.



Model

All three components fitted jointly. This is the example shown on the poster.

Fit (Marginal Type)

χ^2/df	12.6
CV RMSE	0.035
CV rank	1 of 9

Bars: predicted correlation by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line); arrows: CI truncated at 0 or 1.